

AYUSH KASHYAP

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Summary

Full-stack and distributed systems engineer with production experience building **high-throughput backend microservices**, **data pipelines**, and **React/Next.js frontends** in Go, Python, and TypeScript. Hands-on experience designing **agentic AI workflows**, **LLM-powered tools**, and **MCP-style agent orchestration** in production. Active open source contributor to **Ray** and **Milvus** with **7+ merged PRs** — including feature-level C++ database engine work and distributed system bug fixes reviewed by core maintainers. Strong SQL, **distributed data pipeline**, and **low-latency service** fundamentals.

Experience

Software Engineer · Constructure AI

Dec 2025 – Apr 2026 · Remote

- Architected **RESTful microservices** in **Go** and **Python**, reducing **p95 API latency from 1.8s to under 600ms** via Redis caching, connection pooling, and async job queues; built **high-throughput ETL pipelines** processing **500MB+ BIM/IFC datasets** into **PostgreSQL + pgvector** with SQL-based validation, indexing, and query optimization for analytics workloads.
- Designed and shipped a **full-stack NL-to-SQL analytics engine**: **React/Next.js/TypeScript** frontend backed by an **agentic DAG orchestration system** (retrieval → SQL generation → validation → summarization) coordinating **4+ LLM agents**, reducing hallucinated outputs by **40%** and enabling natural language querying of complex construction databases.
- Implemented a **production RAG pipeline** combining **pgvector semantic search** with structured SQL filtering, improving retrieval precision by **28%**; deployed and monitored services across staging and production with strict SLA requirements.

Open Source Contributions

Ray (ray-project/ray)

3+ merged PRs · [PR #63952](#) · [PR #64546](#)

- Patched a **critical distributed systems bug** in Ray Core's C++ **OOM killer**: `TimeBasedWorkerKillingPolicy` killed the `ServeController` under memory pressure, triggering an infinite crash loop; plumbed an `is_system_actor` guard across **protobuf → C++ raylet → Cython → Python** with cross-boundary unit tests.
- Merged [PR #64546](#) contributing to **KV-cache-aware LLM request routing** in Ray Serve — prefix-aware routing reduces redundant GPU recomputation across replicas, directly improving inference throughput for large-scale LLM deployments.

Milvus (milvus-io/milvus)

5+ merged PRs · [PR #51051](#) · [PR #50666](#) · [PR #51636 \(open\)](#)

- Implemented a new **Decimal scalar field type** end-to-end: protobuf wire format, **Go** schema parser, **C++ segcore** storage/indexing/query execution, and filter expression arithmetic — **1000+ line size/XXL PR**, all CI passing, under maintainer review.
- Extended the **filter expression engine** to support bitwise AND, OR, and XOR operators; modified full pipeline — protobuf, ANTLR parser, **C++ bitset engine**, and JSON/array/index execution paths (**500+ lines across 8 files**).

Projects

GoChess Engine

[Demo](#)

- Built a chess engine in **Go** with **Minimax + Alpha-Beta Pruning** (60% game-tree reduction, depth-5 in under 200ms); cross-compiled to **WebAssembly** with **Web Workers** for **2–3×** browser throughput; integrated **Zobrist Hashing**, **Transposition Tables** (100K+ cached positions), **Quiescence Search**, and **PST-based** board evaluation.

PyGit — Distributed Version Control Engine

[GitHub](#)

- Re-implemented **Git internals** (Blob, Tree, Commit) with **content-addressable storage**, **SHA-1 persistence**, **O(1)** object lookup, **staging area**, and **commit DAG traversal**; CLI replicates **init**, **add**, **commit**, **log** with full object serialization and compression.

Technical Skills

Languages: Go, TypeScript, JavaScript, Python, C++, Java, C#

Frontend: React, Next.js, React Native, WebAssembly (WASM), Web Workers

Backend & Distributed Systems: REST APIs, Microservices, PostgreSQL, Redis, ETL Pipelines, Async Processing, Distributed Systems, High-Throughput Services, SQL Optimization

Agentic AI & ML: LangChain, LangGraph, RAG, OpenAI SDK, MCP-style Agent Orchestration, pgvector, Milvus, Prompt Engineering, Multi-Agent Systems

DevOps & Infra: Git, Docker, CI/CD, Ray, Performance Profiling, Observability

Education

Punjab Engineering College (PEC), Chandigarh

B.Tech in Mechanical Engineering

2023 – 2027

CGPA: 8.23 / 10